

CL-1002

Programming Fundamentals

Lab # 7

Objectives:

1. Practice and understanding on basic c++ programs
2. Control Structure
3. If statement
4. If else statement
5. If else-if else statement
6. Data Type Conversion
7. Nested if

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. First think about statement problems and then write your program.
2. Write Program in C compiler/IDE and save source file **for each program**.
3. **Do not copy from any source otherwise you will be penalized with negative marks.**
4. Complete your lab **within given Time Slot**.
5. Add your source code in this word document + Make one ZIP file of your all source codes.
6. Please submit your **Both files** with this naming convention ROLLNO_SECTION_LABNO.
7. Submit your lab on Google Classroom.

Predict the Output of the following code snippet:

Program 1:

```
int main() {  
    int a = 9;  
    if(a>10)  
        cout<<" Number is greater";  
        cout<<" Program End";  
    return 0;  
}
```

Program 2:

```
int main() {  
    int a =3;  
    float b = 3.0;  
    if(a==b)  
        cout << "Both are equal";  
    return 0;  
}
```

Program 3:

```
int main() {  
    char ch = 'A';  
    int a = 65;  
    if(ch==a)  
        cout << "Both are equal";  
    return 0;  
}
```

Program 4:

```
int main() {  
    int a =3;  
    float b = 4.0;  
    if(a==b);  
        cout << "Both are equal";  
    return 0;  
}
```

**Problem: 1 (If- else)****(Marks = 2)**

Write a program that obtains marks in entry test. If obtained marks are greater than 80 then display "You can get admission is Fast" otherwise "Sorry you did not get admission. Please try next time"

Note: Use if else without curly brackets.

Task#02: Datatype Conversion (25 minutes) 4 Marks

Write a C++ program, which asks the user to input a character, and displays its ASCII value (integer).

Write a C++ program, which asks the user to input three floating point values and do the following.

1. First, convert them into integers then display their average as floating-point value?
2. Convert the average value into integer and display its corresponding character?

Hint: Use `static_cast<int>(double)` method or `int(double)` etc

Problem: 3 (if-else if-else)**(Marks = 2)**

Write a computer program that take angle between (0-360) from user and determine the exact quadrant of the angle in the Cartesian plan and also tell the state of both "X" and "Y" axis. For example, if user enter the 95° then is should display "II Quadrant on -XY Axis".

Problem: 4 (If- else if- else)**(Marks = 2)**

Suppose there is a cricket match between Pakistan and India. India played first and gave a target of 300 runs to Pakistan. Now write a program that obtains a cricket match score and remaining wickets from user. If the user is greater than 300 outputs a message indicating "Pakistan win the match by X wickets" while X are remaining wickets. Otherwise shows a message how many remaining runs required to win with X wickets in hand. Use single if-else statement. If score is 300 with 0 wickets in hand then display match is draw.

Note: Score and wickets are not less then zero.

Problem: 5 (If- else if -else)

(Marks = 2)

Write a computer program that charges income tax to different employees of organization named "SpaceX". If employee is in faculty, he/she is **charged 4%** of his salary as income tax. If employee is in management staff, he/she is **charged 3%** of his salary as income tax. If employee is an attendant/guard he/she is **charged 2%** of his salary as income tax.

You need to get salary from the user and then need to calculate the income tax and display the both net salary with income tax of user according to his department.

Note: Use 'F' for faculty, 'M' for Management Staff and 'A' for attendant/guards.

Problem: 6 (nested if-else)

(Marks = 2)

Write a program to calculate the salary as per the following table:

Gender	Years of Service	Qualifications	Salary
Male	≥ 10	Post-Graduate	15000
	≥ 10	Graduate	10000
	< 10	Post-Graduate	10000
	< 10	Graduate	7000
Female	≥ 10	Post-Graduate	12000
	≥ 10	Graduate	9000
	< 10	Post-Graduate	10000
	< 10	Graduate	6000

Note: Need to use Character for Gender ('F', 'M') & Qualification 'P' for Post-Graduate and 'G' for Graduate.

Problem: 7 (if with logical operators)

(Marks = 2)

A program uses a char variable named membership and an int variable named age. The membership variable contains one of the following letters (entered in either uppercase or lowercase): M or N. The letter M stands for *member*, and the letter N stands for *non-member*. The program should

display the appropriate seminar fee, which is based on a person's membership status and age. The fee schedule is shown in Figure. Write the C++ code to display the fee. (Use if-else and nested if-else to solve the problem)

\$10	Club member less than 65 years old
\$5	Club member at least 65 years old
\$20	Non-member

Problem: 8 (using nested If-else statement)

(Marks = 4)

Write a program to find out the roots of quadratic equation ($ax^2 + bx + c = 0$).

The user enter values of a, b and c only. The values entered must be greater than 0, if any of the value is 0 then print "The value can't be zero". If any value entered is less than zero then print "Values must be greater than Zero".

The roots are calculated as:

i) If disc. is greater than zero then roots are real and unequal.

$$\text{root1} = (-b + (\text{discs})^{1/2} / 2 * a).$$

$$\text{root2} = (-b - (\text{discs})^{1/2} / 2 * a).$$

ii) If disc. is less than zero then roots are imaginary .

$$\text{root1} = (-b + i(\text{discs})^{1/2} / 2 * a).$$

$$\text{root2} = (-b - i(\text{discs})^{1/2} / 2 * a).$$

iii) If discs. is equal to zero then roots are real.

$$\text{root1} = \text{root2} = -b / (2 * a).$$

Note:

- "disc" mean Discriminator having value $\text{discs} = b^2 - 4 * a * c$.
- Use `<math.h>` library in your code and use 'sqrtf' for square root.
- The value of iota is constant i.e. $i = -1$.

